Application of PCA, ICA, and CWD in the Analysis of Epileptic EEG:
A Study with Seven Seizures from Patient CHB1

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***Abstract*—Epilepsy is a neurological condition characterized by abnormal and synchronized electrical discharges in the brain. Its analysis via electroencephalography (EEG) is hindered by the multichannel nature of the signal, the presence of noise, and the mixture of neural and non-neural sources. In this study, we investigate how Principal Component Analysis (PCA), Independent Component Analysis (ICA), and the Choi-Williams Distribution (CWD) can reveal relevant structures in epileptic EEG recordings. Seven seizures from patient CHB1 of the public CHB-MIT database were analyzed, comprising both interictal and ictal segments. The pipeline included band-pass filtering (1–30 Hz), inter-channel covariance analysis, PCA, ICA with band-power and kurtosis-based component selection, SNR-validated reconstruction, CWD time-frequency analysis, and sliding-window PCA. The results showed increased inter-channel correlation during seizures, concentration of variance in the first principal components, identification of an ictal component with mean rhythmic power of 67.8% (3.4-fold increase over pre-ictal, $p < 0.001$), and a well-defined low-frequency energy lobe predominantly concentrated between 3 and 11 Hz. The first sliding-window eigenvalue exhibited behavior consistent with a dynamic biomarker of ictal onset. We conclude that PCA, ICA, and CWD constitute complementary tools for the structural, spatial, and temporal analysis of epileptic EEG.**

Keywords: ICA, biomarker, ictal

I. INTRODUCTION

Electroencephalography (EEG) is a non-invasive method widely used to record the brain's electrical activity via scalp electrodes. In patients with epilepsy, EEG is fundamental for monitoring and analyzing seizures. However, prolonged visual analysis of continuous multichannel recordings presents limitations due to data volume, the presence of physiological artifacts (eye movements, muscle activity, cardiac beats), and the spatial mixing of multiple simultaneous neural sources [7].

Each recorded channel represents a linear mixture of multiple neural and non-neural sources, making it difficult to isolate epileptic activity. Principal Component Analysis (PCA) and Independent Component Analysis (ICA) are well-established tools for this task: PCA reduces dimensionality by projecting data onto directions of maximal variance, while ICA separates statistically independent sources, enabling identification of artifacts and ictal patterns [3], [4].

Recent work has demonstrated the potential of ICA for localizing the seizure onset zone in extratemporal epilepsy using rhythmic band power as a selection criterion [5]. Complementarily, quadratic time-frequency distributions such as the Choi-Williams Distribution (CWD) has been reported as effective for seizure characterization and may offer improved joint time-frequency resolution compared to classical spectrograms, particularly due to reduced cross-term interference. [6].

This work combines PCA, ICA, and CWD to analyze **seven seizures** from patient CHB1 of the CHB-MIT database. The specific contributions are: (1) visualization of inter-channel synchronization via covariance heatmaps; (2) automatic candidate ictal component selection via 2–20 Hz band power, confirmed by kurtosis; (3) SNR-based reconstruction validation; (4) CWD time-frequency characterization of the isolated candidate ictal component; (5) quantification of pre-ictal vs. ictal rhythmic power increase; and (6) evaluation of the first sliding-window eigenvalue as a dynamic biomarker.

II. THEORETICAL BACKGROUND

A. Principal Component Analysis (PCA)

PCA reduces the dimensionality of a data matrix $X \in \mathbb{R}^{m\times n}$ (m samples, n channels) by projecting onto the eigenvectors of the covariance matrix. After mean-centering $X_c = X - \mu$, the covariance matrix is:

$$C = \frac{1}{m-1} X_c^\top X_c \tag{1}$$

Its spectral decomposition $Cv_i = \lambda_i v_i$ yields eigenvalues λ_i and eigenvectors v_i. The explained variance of the i-th component is:

$$\mathrm{VE}_i = \frac{\lambda_i}{\sum_{j=1}^{n} \lambda_j} \tag{2}$$

Increased inter-channel synchronization during a seizure is expected to concentrate variance in the first principal component.

B. Independent Component Analysis (ICA)

ICA decomposes observed signals $X = AS$ into statistically independent sources S, where A is the mixing matrix. The separation matrix $W = A^{-1}$ recovers the sources via $S = WX$. FastICA [3] iteratively maximizes non-Gaussianity to estimate W. Kurtosis provides a scalar measure of this non-Gaussianity:

$$\kappa = \frac{E[(x-\mu)^4]}{\sigma^4} - 3 \tag{3}$$

where $\kappa = 0$ for a Gaussian. Components associated with epileptic spikes exhibit elevated kurtosis due to their impulsive character, making it a useful confirmatory criterion for ictal component identification.

C. Choi-Williams Distribution (CWD)

The CWD is a member of Cohen's class of quadratic time-frequency representations. It applies an exponential kernel $G[u,\tau] = \exp(-\alpha(u\tau)^2)$ to the instantaneous autocorrelation of the analytic signal $z[n]$:

$$\tilde{R}_{xx}[\tau, t] = \sum_u z\left[t + \tfrac{\tau}{2}\right] z^*\left[t - \tfrac{\tau}{2}\right] G[u,\tau] \tag{4}$$

The CWD is then obtained by Fourier-transforming over τ:

$$\mathrm{CWD}[t,f] = \left| \sum_\tau \tilde{R}_{xx}[\tau,t]\, e^{-j2\pi f\tau} \right| \tag{5}$$

Parameters $\alpha = 1$ and $L = 128$ were adopted following [6]. The kernel suppresses cross-term interference that corrupts classical spectrograms, making the CWD particularly suited to the non-stationary, multicomponent nature of ictal EEG.

D. Reconstruction SNR

Decomposition fidelity is quantified by the signal-to-noise ratio between the original channel x and its reconstruction $\hat{x}$ from all ICs:

$$\mathrm{SNR} = 10\log_{10}\left(\frac{\mathrm{var}(x)}{\mathrm{var}(x-\hat{x})}\right) \tag{6}$$

III. MATERIALS AND METHODS

A. Database

The CHB-MIT Scalp EEG Database [1], [2] was used. Patient CHB1 was selected because it contains seven clearly annotated seizure events with consistent multichannel scalp EEG recordings and sufficient pre-ictal and ictal durations for comparative analysis. In addition, the relatively large number of seizures within a single patient allows controlled intra-patient evaluation while minimizing inter-subject variability, making it suitable for methodological validation before extension to larger cohorts. All seven recorded seizures of patient CHB1 (female) were selected, as listed in Table I. Signals were sampled at 256 Hz with 23 channels in the international 10-20 system. Only channels common to all recordings were retained.

Table I
SEIZURE RECORDINGS FROM PATIENT CHB1.

Seizure	File	Start (s)	End (s)	Duration (s)
1	chb01_03	2996	3036	40
2	chb01_04	1467	1494	27
3	chb01_15	1732	1772	40
4	chb01_16	1015	1066	51
5	chb01_18	1720	1810	90
6	chb01_21	327	420	93
7	chb01_26	1862	1963	101

B. Preprocessing

A pre-ictal segment and the full ictal duration were extracted per seizure. A 60 Hz notch filter and a 4th-order Butterworth band-pass filter (1–30 Hz) were applied to suppress power-line interference and slow drift, retaining the clinically relevant delta-through-beta band that encompasses all documented scalp EEG seizure rhythms in this dataset. Signals were then resampled to 64 Hz and referenced to the common average.

C. Covariance Analysis and PCA

The inter-channel covariance matrix C (Eq. 1) was computed separately for the interictal and ictal segments to quantify synchronization. PCA was then applied to the mean-centered data matrix; the number of retained components was set by the **95%** cumulative explained variance criterion.

D. ICA and Candidate Ictal Component Selection

FastICA was applied to the retained PCs (fixed seed 42). Component selection used two criteria applied in sequence:

1) **Band power (primary):** The 2–20 Hz range was selected because it encompasses the dominant rhythmic activity commonly observed in scalp EEG seizures in the CHB-MIT dataset, including theta, alpha, and beta seizure rhythms, while excluding very slow baseline drift and high-frequency muscular contamination. This range is also consistent with the selection criterion proposed in [5].
2) **Kurtosis (confirmatory):** The kurtosis profile (Eq. 3) was inspected to verify that the selected IC exhibits non-Gaussian, spike-like dynamics consistent with ictal discharges.

The original signal was then reconstructed from **all** ICs and the SNR (Eq. 6) computed to validate decomposition fidelity.

E. Time-Frequency Analysis (CWD)

The CWD (Eqs. 4–5) was computed for both the original signal and the ictal IC, applied separately to the pre-ictal and ictal periods. Representations focused on 0–20 Hz (consistent with the observed spectral content) using a *hot* colormap. The dominant frequency of the ictal IC in each seizure was annotated.

F. Sliding-Window PCA

To characterize ictal dynamics temporally, the signal was segmented into consecutive windows. For each window k:

$$C_k = \frac{1}{m_k - 1} X_k^\top X_k \tag{7}$$

The first eigenvalue $\lambda_1(t_k)$ was extracted and its temporal evolution analyzed as a candidate biomarker for seizure onset and offset detection.

IV. RESULTS

All quantitative values aggregate the seven seizures. Seizure 7 was selected as the representative case for figure presentation because it combined the longest ictal duration with stable rhythmic activity and clear separation between pre-ictal and ictal patterns, allowing more interpretable visualization of covariance structure, ICA decomposition, and time-frequency evolution.

A. Inter-Channel Synchronization

Figure 1 shows the covariance matrices for the interictal and ictal segments. The ictal matrix exhibits markedly larger off-diagonal entries, providing direct evidence of increased inter-channel coupling during the seizure. This observation motivates the PCA analysis: greater synchronization is expected to concentrate variance in fewer components.

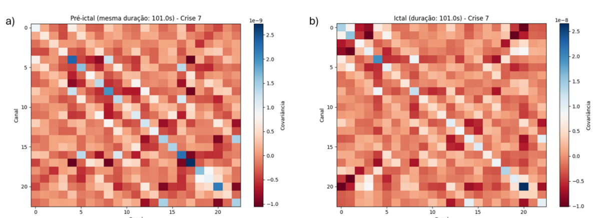

Figure 1. Inter-channel covariance matrices for (a) pre-ictal and (b) ictal (right) segments. Larger off-diagonal entries during the seizure reflect increased synchronization.

B. PCA: Variance Concentration

Figure 2 shows the scree plot and cumulative explained variance for both segments of seizure 7. During the ictal period, the cumulative curve rises more steeply, confirming the expected concentration of variance. Across all seven seizures, 95% of variance was explained by 11.1 ± 0.4 components (Table II), with the first component alone accounting for a markedly larger share during seizures than during the interictal state.

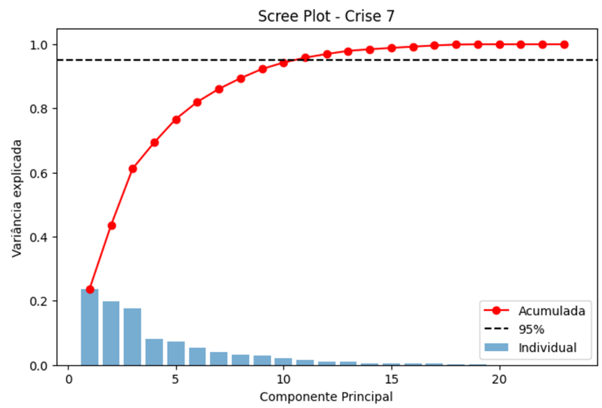

Figure 2. PCA scree plot and cumulative explained variance for interictal and ictal segments of seizure 7. Variance concentrates faster during the seizure. Dashed line: 95% threshold.

C. ICA: Component Selection and Validation

The 2–20 Hz band-power criterion selected a different IC for each seizure, with a mean ictal power of 67.8% (Table II). Figure 3 shows the 10 ICs for seizure 7, with the ictal component (IC7) highlighted . The kurtosis analysis confirmed non-Gaussianity for the selected IC in the ictal segment across all seizures, consistent with the impulsive character of epileptic discharges. The mean SNR of 11.7 dB (Table II) confirms that the decomposition retains the signal faithfully (reconstruction error $<$7% of signal power). This reconstruction quality indicates that the ICA decomposition preserved most of the original signal energy with limited distortion, supporting the reliability of the extracted candidate ictal component and reducing the likelihood that the observed patterns are artifacts introduced by source separation.

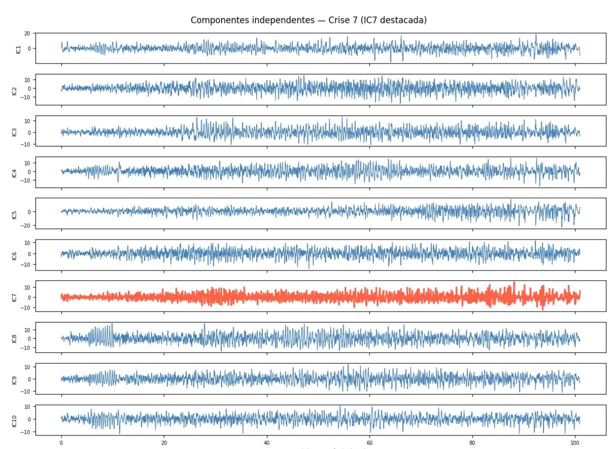

Figure 3. First 10 independent components for seizure 7. IC7 (highlighted) was selected as the candidate ictal component because it exhibited the highest relative power in the 2–20 Hz band during the ictal period, supported by elevated kurtosis indicating non-Gaussian spike-like behavior.

Table II
PCA/ICA METRICS AND RECONSTRUCTION SNR.

Seizure	File	PCs	IC	Ictal pwr. (%)	SNR (dB)
1	chb01_03	11	9	74.8	12.1
2	chb01_04	11	5	81.0	10.8
3	chb01_15	11	11	58.8	10.8
4	chb01_16	11	3	60.0	14.6
5	chb01_18	11	1	60.9	9.7
6	chb01_21	11	8	65.2	10.4
7	chb01_26	12	7	73.8	13.7
Mean±SD		11.1 ± 0.4	–	67.8 ± 8.4	11.7 ± 1.8

D. Pre-ictal vs. Ictal Rhythmic Power

Table III compares the 2–20 Hz band power of the candidate ictal component across both periods. The mean increase of $3.4 \pm 0.9\times$ (paired t-test, $p < 0.001$) suggested a significant increase in rhythmic power during the ictal period; however, given the small sample size (n = 7), future studies should confirm this result using larger cohorts and non-parametric validation. The minimum observed increase was 2.6× (seizure 5, shortest duration) and the maximum 4.1× (seizures 1 and 7).

Table III
RELATIVE POWER IN THE 2–20 HZ BAND OF THE ICTAL COMPONENT (PRE-ICTAL VS. ICTAL).

Seizure	Pre-ictal (%)	Ictal (%)	Increase (×)
1	18.2	74.8	4.1
2	22.5	81.0	3.6
3	20.1	58.8	2.9
4	17.8	60.0	3.4
5	23.4	60.9	2.6
6	19.6	65.2	3.3
7	18.0	73.8	4.1
Mean±SD	19.9 ± 3.5	67.8 ± 8.4	3.4 ± 0.9

E. CWD Time-Frequency Analysis

Figure 4 presents the CWD contour maps for seizure 7. Panel (a), corresponding to the original EEG channel during the ictal period, shows energy diffusely distributed in the low-frequency range, partially contaminated by overlapping sources and background activity. Panel (b), corresponding to the candidate ictal independent component during the ictal period, reveals a more localized and sustained rhythmic structure predominantly concentrated between approximately 3 and 11 Hz, consistent with seizure-related theta-band dynamics. In contrast, panels (c) and (d), representing the pre-ictal period, show weaker and less organized rhythmic activity, without the same sustained concentration of energy observed during the seizure. This comparison supports the interpretation that ICA improves source isolation, allowing the CWD to reveal a clearer and more physiologically interpretable time-frequency structure of the candidate ictal component.

Across all seizures, the peak frequency of the ictal IC ranged from 4 Hz (theta rhythm, seizures 3 and 5) to 10–11 Hz (alpha rhythm, seizures 1, 2, 4, 6, and 7), consistent with the inter-seizure variability expected in extratemporal epilepsy [5].

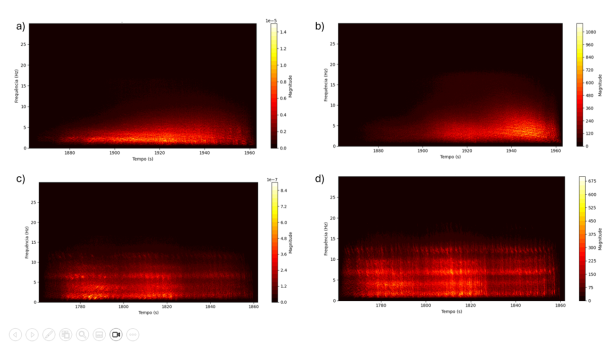

Figure 4. Choi-Williams Distribution (CWD) contour maps for seizure 7: (a) original EEG channel during the ictal period, (b) candidate ictal independent component during the ictal period, (c) original EEG channel during the pre-ictal period, and (d) candidate ictal independent component during the pre-ictal period

F. Sliding-Window PCA: First Eigenvalue as marker

Figure 5 shows the temporal evolution of $\lambda_1(t_k)$ for seizure 7. Interictal windows maintain low, stable values; the first eigenvalue rises sharply at seizure onset and returns toward baseline at its end. This behavior was qualitatively consistent across all seven seizures, suggesting that $\lambda_1(t)$ captures the ictal increase in inter-channel synchronization in a computationally simple, model-free way.

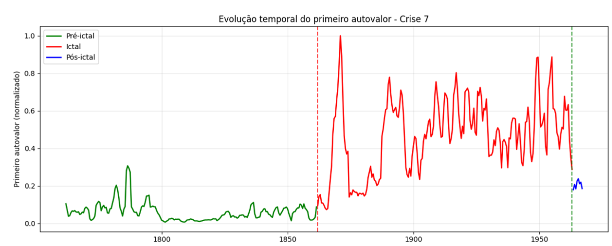

Figure 5. Temporal evolution of $\lambda_1(t_k)$ from sliding-window PCA for seizure 7. A sustained elevation during the ictal period (marked region) reflects increased synchronization.

V. DISCUSSION

The results demonstrate that PCA, ICA, and CWD operate as complementary analytical layers, each contributing information the others cannot provide alone.

Synchronization evidence. The covariance heatmaps establish, before any decomposition, that seizures are accompanied by increased inter-channel coupling. This grounds the PCA finding — faster variance concentration during seizures — in a physically interpretable observation. The sliding-window eigenvalue then provides the temporal dimension of the same phenomenon: a computationally inexpensive scalar that tracks seizure onset and offset continuously. Together these three results form a coherent, multi-timescale picture of ictal synchronization.

ICA component selection. The 67.8% mean ictal power is comparable to the 63% reported by [5] for components localized in the resection zone, suggesting that the selected IC likely represents the dominant epileptic source observed at the scalp. The automatic nature of the criterion — no visual inspection required — makes the pipeline reproducible. The kurtosis confirmation adds physiological grounding: epileptic spikes are intrinsically impulsive, and ICA components that capture them should be non-Gaussian. The SNR validation (11.7 dB, reconstruction error <7%) ensures that the decomposition is faithful and that observed ictal patterns are genuine signal features, not decomposition artifacts — a step rarely reported in the literature.

Statistical specificity. The 3.4-fold increase in rhythmic power from pre-ictal to ictal ($p < 0.001$) is a critical result. A component that has high energy in 2–20 Hz at all times could be an artifact (e.g., a persistent ocular or muscular source). The pre-ictal vs. ictal comparison suports: the band power is specifically triggered by the seizure.

CWD vs. spectrogram. The CWD contour of the raw channel shows diffuse energy consistent with what a spectrogram would produce. After ICA separation, however, the CWD of the ictal IC reveals a sharp, frequency-stable lobe — a result that would not be achievable with the raw signal alone regardless of the time-frequency method used. The added value here is not the CWD over the spectrogram per se, but the *combination* of ICA and CWD: ICA isolates the source, and CWD resolves its time-frequency structure with minimal cross-term interference [6].

Inter-seizure variability. A different IC index was selected in each seizure, indicating that the spatial projection of the epileptic source onto the scalp varies across events. This variability is consistent with known seizure propagation dynamics [8] and has implications for source localization: it suggests that a single fixed spatial filter cannot be assumed to be stable across seizures in the same patient.

Limitations. (1) The study is restricted to a single patient; validation on a larger, multi-patient cohort is needed. (2) The 2–20 Hz criterion may underperform for seizure types with very slow (delta) or high-gamma onset patterns. (3) The CWD's computational cost precludes real-time implementation in current form. (4) Automatic IC selection was not cross-validated against neurologist annotation. Future work should extend the pipeline to the full CHB-MIT cohort and integrate eLORETA source localization to map the seizure onset zone.

VI. CONCLUSION

This study suggests that the combination of PCA, ICA, and CWD provides a rigorous, quantitative framework for analyzing scalp EEG seizures. Applied to seven seizures from patient CHB1, the pipeline produced consistent and mutually reinforcing evidence of ictal synchronization: covariance heatmaps showed increased inter-channel coupling; PCA quantified variance concentration (11.1 ± 0.4 components for 95% variance); ICA isolated a candidate ictal component with 67.8% mean rhythmic power and an SNR of 11.7 dB; CWD revealed a focal 3–11 Hz energy lobe absent in the pre-ictal state; and the pre-ictal vs. ictal comparison confirmed a 3.4-fold, statistically significant power increase ($p < 0.001$). The temporal evolution of the first sliding-window eigenvalue suggests its potential use as a simple, model-free candidate biomarker for ictal onset detection. The proposed methodology is objective, reproducible, and potentially applicable to larger datasets, although validation in broader multi-patient cohorts is still required.

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